

PORTLAND HILLSBORO, OR, USA

WMO: 726986

Lat: 45.541N

Lon: 122.949W

Elev: 62

StdP: 100.58

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94261

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -4.9	(c) -3.0	(d) -11.3	(e) 1.4	(f) -1.1	(g) -7.8	(h) 1.9	(i) 0.7	(j) 10.4	(k) 6.8	(l) 9.0	(m) 5.5	(n) 1.4	(o) 310	(p) 0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.3	(c) 33.6	(d) 19.8	(e) 31.3	(f) 19.2	(g) 28.9	(h) 18.3	(i) 20.9	(j) 30.9	(k) 19.9	(l) 29.6	(m) 19.0	(n) 27.8	(o) 3.2	(p) 60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	12.5	24.4	16.3	11.7	22.8	15.4	11.0	21.8	60.4	31.1	56.9	29.5	53.7	28.0	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.2	(b) 7.3	(c) 6.1	(e) -8.4	(f) 37.4	(g) 2.9	(h) 2.0	(i) -10.5	(j) 38.9	(k) -12.2	(l) 40.1	(m) -13.8	(n) 41.2	(o) -15.9	(p) 42.7		
(a) 8.2	(b) 7.3	(c) 6.1	(e) -8.9	(f) 23.6	(g) 2.9	(h) 2.1	(i) -11.0	(j) 25.0	(k) -12.7	(l) 26.2	(m) -14.3	(n) 27.4	(o) -16.3	(p) 28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	4.8	5.8	8.0	10.2	13.7	16.3	19.7	19.7	16.7	11.5	7.2	4.4
	DBStd	6.27	3.45	3.07	2.85	3.08	3.37	3.23	3.09	2.85	3.13	2.96	3.45	3.49
	HDD10.0	692	163	119	77	33	6	0	0	0	1	19	97	177
	HDD18.3	2649	419	351	322	246	148	78	19	17	68	213	335	433
	CDD10.0	1245	2	2	13	38	121	189	300	299	201	64	12	2
	CDD18.3	161	0	0	0	1	5	17	61	59	18	1	0	0
	CDH23.3	2637	0	0	1	28	143	315	912	848	374	16	0	0
	CDH26.7	1066	0	0	0	6	39	121	399	365	134	2	0	0
Wind	WSAvg	2.2	2.1	2.3	2.4	2.4	2.2	2.2	2.3	2.1	1.9	1.8	2.0	2.2
Precipitation	PrecAvg	1104	174	112	112	81	63	41	11	17	35	95	168	191
	PrecMax	1716	359	275	197	170	132	96	44	69	159	224	397	400
	PrecMin	824	66	6	38	30	0	6	0	0	0	16	57	57
	PrecStd	211	67	58	43	38	35	20	10	15	31	49	78	85
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	16.2	21.3	27.1	30.3	34.2	37.1	36.2	33.4	25.7	17.5	14.2
		MCWB	12.3	10.6	12.9	16.8	18.3	20.1	21.1	21.1	18.8	15.6	13.8	12.4
	2%	DB	12.5	14.0	17.9	22.6	27.4	30.2	33.7	33.2	30.4	22.2	15.6	12.3
		MCWB	10.5	10.0	11.6	14.6	16.8	18.9	20.1	19.5	17.7	14.3	13.0	11.0
	5%	DB	11.1	12.4	15.3	19.1	24.1	27.1	31.1	30.9	27.7	19.5	13.9	10.9
		MCWB	9.4	9.1	10.2	12.5	15.6	17.7	19.6	19.2	17.3	13.2	11.6	9.7
	10%	DB	9.9	11.0	13.3	16.4	21.2	23.9	28.3	28.0	25.0	17.3	12.5	9.1
		MCWB	8.4	8.4	9.7	11.1	14.4	16.2	18.6	18.4	16.4	13.0	10.6	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	13.2	14.1	17.8	19.6	21.6	22.8	22.6	20.6	18.0	15.3	12.8
		MCDB	14.2	14.3	19.2	25.4	28.9	31.5	33.5	32.0	30.1	22.5	16.6	13.7
	2%	WB	11.1	11.0	12.4	15.0	17.7	19.5	21.0	20.7	19.0	15.8	13.7	11.1
		MCDB	12.0	12.8	15.8	21.4	25.1	29.0	31.0	30.7	27.7	19.2	15.0	12.1
	5%	WB	9.8	9.8	11.3	13.1	16.3	17.9	20.0	19.7	17.9	14.6	12.3	9.7
		MCDB	10.7	11.7	14.3	17.9	22.9	25.8	29.9	29.0	25.6	17.8	13.4	10.6
	10%	WB	8.6	8.6	10.3	11.7	14.9	16.6	18.9	18.7	16.9	13.6	11.0	8.1
		MCDB	9.6	10.4	13.0	15.6	20.5	23.3	27.9	27.4	23.4	16.4	12.1	8.9
Mean Daily Temperature Range	5% DB	MDBR	6.3	8.3	9.5	10.7	11.8	12.5	15.3	15.5	14.6	10.9	7.7	6.1
		MCDBR	7.3	10.8	13.8	17.2	18.3	18.7	20.2	20.4	20.1	16.2	9.2	6.5
		MCWBR	5.3	7.0	8.0	9.5	9.4	9.1	8.6	8.8	9.3	9.1	6.2	5.3
	5% WB	MCDBR	6.1	8.1	11.3	15.0	16.0	17.1	18.5	18.4	17.4	11.7	7.0	6.0
		MCWBR	5.0	5.8	7.1	8.7	8.6	8.7	8.7	8.6	8.6	7.3	5.2	5.2
Clear-Sky Solar Irradiance	taub	0.320	0.316	0.330	0.358	0.372	0.361	0.349	0.354	0.342	0.331	0.323	0.325	
	taud	2.531	2.553	2.520	2.404	2.383	2.426	2.487	2.485	2.508	2.546	2.530	2.477	
	Ebn at Noon	794	874	906	902	896	904	912	895	880	841	781	739	
	Edh at Noon	66	77	90	110	117	112	104	101	91	76	65	63	
All-Sky Solar Radiation	RadAvg	1.16	2.00	2.91	4.24	5.40	5.97	6.80	5.81	4.34	2.51	1.35	0.95	
	RadStd	0.19	0.28	0.39	0.40	0.52	0.56	0.47	0.39	0.38	0.32	0.18	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-0.48	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+23

Nomenclature: See separate page